Problem 4.9

Suppose that the accumulation function for an account is $a(t) = 1+3it$. At time 0, you invest \$100 in this account. If the value in the account at time 10 is \$420, what is i?

Solution

Since

$$a(10) = \frac{A(10)}{A(0)},$$

and $A(10) = 420$, $A(0) = 100$, we have

$$a(10) = \frac{420}{100} = 4.20.$$

But the accumulation function is

$$a(t) = 1 + 3it.$$

So at $t = 10$,

$$1 + 30i = 4.20.$$

Therefore,

$$30i = 3.20,$$

and hence

$$i = \frac{3.20}{30} = 0.1066\overline{6}.$$

Thus,

$$i \approx 10.66\%.$$